**Supplementary Table S1:** L

| Authors | Year | Country | Patient | Age(years) | Phenotypes |
|---|---|---|---|---|---|
| **Schwartz et al. [3]** | 2000 | USA§ | 1(FC) | 7 | Brachydactyly/wide fingers; Cupid's Bow; Dorsal Groove penis; Downward Slant Palpebral Fissures; Hypertelorism ; Interphalangeal Joint Contracture; Joint Hypermobility; Long Philtrum; Macrocephaly; Pointed Chin.; Ptosis; Short Stature; Simian Creases; Umbilical Hernia |
| | | | 2(IT) | 12 | Brachydactyly/wide fingers; Cupid's Bow; Downward Slanted Palpebral Fissures; Hypertelorism; InguiNA†l Hernia; Interphalangeal Joint Contracture; Joint hypermobility; Long Philtrum; Macrocephaly; Pointed Chin; Ptosis; Shawl Scrotum; Short Stature; Simian Crease |
| | | | 3(MK) | 11 | Anteverted Nostrils; Brachydactyly/Wide Fingers; Hypertelorism; Shawl Scrotum; Short Stature; Syndactyly; Cryptorchidism; Downward slant palpebral fissures; InguiNA†l hernia; Ptosis; Widow's peak; Bone Age retarded; Myopia |

| | | | | | |
|---|---|---|---|---|---|
| **Orrico et al. [37]** | 2000 | Italy | 4(II1) | 66 | Anteverted Nostrils; Camptodactyly; Clinodactyly; Crase Below Lower Lip; Cryptorchidism; Downward Slanted Palpebral Fissures; Hypertelorism; InguiNA†l Hernia; Interdigital Webbing; Ptosis; Shawl Scrotum; Short Stature; Short/Broad Hands; Umbilical Hernia; Widow's Peak |
| | | | 5(IV9) | 9 | Anteverted Nostrils; Anteverted nostrils/Short Nose; Clinodactyly; Crease Below Lower Lip; Cryptorchidism; Downward Slant Palpebral Fissures; Dysplastic Ears; Frontal Bossing; Hypertelorism; InguiNA†l Hernia; Interdigital Webbing; Joint Hypermobility; Long Philtrum; Midface Hypoplasia; Ptosis; Round Face; Shawl Scrotum; Short Stature; Short/Broad Hands; Umbilical Hernia; Wide feet; Widow's peak |
| | | | 6(IV10) | 19 | Anteverted Nostrils; Anteverted nostrils/Short Nose; Clinodactyly; Crease Below Lower Lip; Cryptorchidism; Downward Slant Palpebral Fissures; Hypertelorism; InguiNA†l Hernia; Interdigital Webbing; Joint Hypermobility; Ptosis; Shawl Scrotum; Short Stature; Short/Broad Hands; Umbilical Hernia; Widow's Peak |

| | | | | | |
|---|---|---|---|---|---|
| | | NR* | 7(50) | 29 | Anteverted Nostrils; Anteverted nostrils/Short Nose; Brachydactyly/Wide Fingers; Camptodactyly; Clinodactyly; Cryptorchidism; Hydrocele; Hypertelorism; InguiNA†l Hernia; Joint Hypermobility; Long Philtrum; Oligoasthenozoospermia; Ptosis; Shawl Scrotum; Short Stature; Syndactyly |
| | | NR* | 8(90) | 9 | Anteverted Nostrils; Brachydactyly/Wide Fingers; Camptodactyly; Cryptorchidism; Downward Slant Palpebral Fissures; Dysplastic Ears; Frontal Bossing; Hypertelorism; InguiNA†l Hernia; Interdigital Webbing; Joint Hypermobility; Long Philtrum; Ptosis; Round Face; Shawl Scrotum; Short Stature; Short/Broad Hands; Syndactyly; Widow's Peak |
| | | Portugal | 9(25) | 11 | Anteverted Nostrils; Brachydactyly/Wide Fingers; Camptodactyly; Cryptorchidism; Downward Slant Palpebral Fissures; Dysplastic Ears; Frontal Bossing; Hypertelorism; InguiNA†l Hernia; Interdigital Webbing; Joint Hypermobility; Long Philtrum; Ptosis; Round Face; Shawl Scrotum; Short Stature; Short/Broad Hands; Syndactyly; Widow's Peak |

| | | | | | |
|---|---|---|---|---|---|
| **Orrico et al. [6]** | 2004 | NR* | 10(53) | 1.3 | Anteverted nostrils/Short Nose; Brachydactyly/wide fingers; Clinodactyly; Cryptorchidism; Downward slant palpebral fissures; Hypertelorism; InguiNA†l hernia ; Joint hypermobility; Long Philtrum;Shawl Scrotum ;Short stature; Simian Crease; Syndactyly; Umbilical hernia; Widow’s peak |
| | | NR* | 11(61) | 21 | Behavioral Disorders;Brachydactyly/wide fingers; Camptodactyly; Hypertelorism; Joint hypermobility;Ptosis; Shawl Scrotum; short neck; Short stature; Social/Language skills delayed; Strabismus; Syndactyly |
| | | NR* | 12(26) | 3 | Anteverted nostrils/Short Nose;Anteverted nostrils/Short Nose; Brachydactyly/wide fingers; Clinodactyly; Epicanthal fold; Hypertelorism ; InguiNA†l hernia; Joint hypermobility; Long philtrum; Metatarsus varus; Round face; Shawl Scrotum; Short stature;Syndactyly; Umbilical hernia; Widow’s peak |
| | | NR* | 13(73) | 4.5 | Anteverted nostrils/Short Nose; Bone age retarded; Camptodactyly; Cleft Lip Palate; Downward slant palpebral fissures; Epicanthal fold; Hypertelorism; InguiNA†l Hernia; Joint Hypermobility; Long Philtrum; Low set ears; Ptosis; Shawl Scrotum; Syndactyly; Widow's peak |

| | | | | | |
|---|---|---|---|---|---|
| | | NR* | 14(58) | 1.6 | ADHD; Anteverted nostrils/Short Nose; Axial Hypotonia;Brachydactyly/Wide fingers; Cryptorchidism; Development delayed; Hypertelorism; Joint Hypermobility; Long Philtrum; Low set ears; Prominent umbilicus; Ptosis; Shawl Scrotum; Short Stature; Syndactyly |
| | | Ireland | 15(65) | 16 | Anteverted nostrils/Short Nose;Brachydactyly/Wide Fingers; Camptodactyly; Cryptorchidism; Downward slant palpebral fissures; Hypertelorism; Long philtrum; Ptosis; Shawl Scrotum; Short/Broad Hands; Syndactyly; Wide feet |
| | | | | | |
| **Orrico et al. [7]** | 2005 | Italy | 16 | 16 | ADHD; Axial hypotonia; Behavioral disorder; Borderline mental retardation;Brachydactyly/Wide fingers; Broad NA†sal bridge; Downward slant palpebral fissures; Dysplastic ears; Enlarged Virchow-Robin spaces; HyperNA†sal speech; Hypertelorism; Interdigital webbing; Joint hypermobility; Low set ears; MicrogNA†thia; Midface Hypoplasia; Ogival palate; Pectus excavatum; Prominent umbilicus; ptosis; Short stature; Simian Creases; Strabismus |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **Shalev et al. [24]** | 2006 | Israel | 17(IV2) | 24 | Anteverted nostrils/Short Nose; Brachydactyly/Wide fingers; Camptodactyly; Dysplastic ears; Hypertelorism; Long philtrum; Pectus cariNA†tum; Ptosis; Round face; Shawl Scrotum Short stature; Simian creases; Syndactyly |
| | | | 18(IV3) | 16 | Brachydactyly/wide finger; Camptodactyly; Dysplastic ears; Frontal bossing; Hypertelorism; Ptosis; Round face; Short Stature; Simian creases; Syndactyly |
| | | | 19(IV4) | 2 | Anteverted nostrils/Short Nose; Behavioral disorder; Development delayed; Fine and gross motor skills delayed; Hypertelorism; Microcephaly, Neurocognitive development delayed; Ptosis; Seizures; Short stature; Social/Language Skills delayed; Vertigo |
| | | | | | |
| **KaNA†me et al. [8]** | 2006 | Japan | 20(1) | 13 | ADHD;Brachydactyly/Wide fingers; Frontal bossing; Hypertelorism; Interdigital webbing; Interphalangeal joint contracture; Joint hypermobility; Ptosis; Shawl scrotum; Short stature; Short/Broad Hands; Wide feet |
| | | | 21(2) | 4 | Brachydactyly/Wide fingers; Frontal Bossing; Hypertelorism; Interdigital Webbing; Interphalangeal joint contracture; Joint hypermobility; Shawl Scrotum; Short stature; Short/Broad Hands; wide feet |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **Bottani et al. [25]** | 2007 | Switzerland | 22 | 8 | Clinodactyly; Clubbed feet;Cryptorchidism; InguiNA†l hernia; Joint hypermobility; Pectus excavatum; Seizures; Shawl Scrotum; Unilateral focal frontoparietal PMG, Migraine headaches |
| | | | | | |
| **Orrico et al. [26]** | 2007 | Italy | 23 | 15 | Agenesis of Corpus Callosum; Anteverted nostrils/Short Nose;Brachydactyly/wide fingers; Crease below lower lip; Downward slant palpebral fissures ; Dysplastic ears; Fine and gross motor skills delayed; Hydrocele; Hypertelorism ; InguiNA†l hernia ; Interdigital webbing; Long philtrum; Midface Hypoplasia; Neurocognitive development delayed; Omphalocele; Parietal-occipital encephalocele; Pectus excavatum; Ptosis; Right hemifacial microsomia; Shawl Scrotum; Short neck; Short stature; Short/Broad Hands; Simian Creases; still's murmur ; Submucous cleft palate; Synophry; Transethmoidal;Meningoencephalocele; Ventricular septal defect; Wide feet;Widow's Peak |
| | | | 24 | 6 | Anteverted nostrils/Short Nose; Bifid uvula; Brachydactyly/Wide fingers; Camptodactyly; Crease below lower lip; InguiNA†l hernia; Interdigital webbing; Long philtrum; Low set ears; Metatarsus varus; Pectus excavatum; Prominent umbilicus; Ptosis; Round face; Shawl Scrotum; Short neck; Short stature; Simian Creases; Synophrys; Telecanthus; Widow's peak |

| | | | | | |
|---|---|---|---|---|---|
| **Bedoyan et al. [27]** | 2009 | USA§ | 25 | 1.25 | Anteverted nostrils/Short Nose; Astigmatism; Camptodactyly; Cryptorchidism; Downward slant palpebral fissures; Dysplastic ears; Hypertelorism; InguiNA†l hernia; Ogival palate; Ptosis; Shawl Scrotum |
| | | | | | |
| | | NR* | 26(9) | NR* | Anteverted nostrils; Brachydactyly/Wide fingers; Clinodactyly; Cryptorchidism; Dental malocclusion; Dysplastic ears; Hypertelorism; long philtrum; Midface hypoplasia; Obesity; Shawl scrotum; Short stature; Short/Broad Hands; Wide feet |
| | | NR* | 27(2) | NR* | Anteverted nostrils/Short Nose; Bone age retarded; Development delayed; Dysplastic ears; Hypertelorism; Obesity; preauricular tag; Shawl scrotum; Short stature; Short/Short/Broad Hands; Simian creases; Widow's peak |
| | | NR* | 28(5) | NR* | Anteverted nostrils/Short Nose;Camptodactyly; Cryptorchidism; Development delayed; Hypertelorism; InguiNA†l hernia; Interdigital webbing; Long philtrum; Metatarsus varus; Ptosis; Shawl scrotum; Short stature; Short/Broad Hands; Simian creases; Umbilical hernia; Wide feet; Widow's peak |

| | | | | | |
|---|---|---|---|---|---|
| **Orrico et al. [16]** | 2010 | NR* | 29(10) | NR* | Anteverted nostrils/Short Nose; Brachydactyly/wide fingers; Camptodactyly; Clinodactyly; Crease below lower lip; Downward slant palpebral fissures; Fetal Nuchal Oedema; Frontal bossing; Hypertelorism; Interdigital webbing; Long philtrum; Metatarsus varus; Microcephaly; Ptosis; Short stature; Short/Broad Hands; Simian creases; Wide feet |
| | | NR* | 30(3) | NR* | Anteverted nostrils/Short Nose; Brachycephaly; Brachydactyly/wide fingers; Clinodactyly; Crease below lower lip; Development delayed; Dysplastic ears; Hypertelorism; Hypospadias; InguiNA†l hernia; Long Philtrum; Micropenis; Midface hypoplasia; Obesity; Ptosis; Shawl scrotum, Short stature; Short/Short/Broad Hands; Umbilical hernia; Widow's peak |
| | | NR* | 31(11) | NR* | Anteverted nostrils/Short Nose; bone age retarded; Brachydactyly/Wide fingers; Clinodactyly; crease below lower lip; downward slant palpebral fissures; frontal bossing; Hypertelorism; long philtrum; midface hypoplasia; Patent ductus arteriosus; prominent umbilicus; ptosis; shawl scrotum; Short Stature;Short/Broad Hands; wide feet; widow's peak |
| | | NR* | 32(8) | NR* | Anteverted nostrils/Short Nose;Anteverted nostrils/Short Nose; Brachydactyly/wide fingers;Camptodactyly; Clinodactyly; Cryptorchidism; Hypertelorism; long philtrum; obesity; Shawl scrotum; Short stature; Short/Broad Hands; Wide feet |

| | | | | | |
|---|---|---|---|---|---|
| | | NR* | 33(6) | NR* | Anteverted nostrils/Short Nose; bone age retarded; Brachydactyly/Wide fingers; Camptodactyly; clinodactyly; crease below lower lip; cryptorchidism; downward slant palpebral fissures; dysplastic ears; hypertelorism; joint hypermobility; shawl scrotum; Short stature; Short/Broad Hands; simian creases; Strabismus; wide feet; widow's peak |
| | | NR* | 34(7) | NR* | Anteverted nostrils/Short Nose; Brachydactyly/wide fingers; Clinodactyly; Congenital glaucoma; crease below lower lip; cryptorchidism; Dysplastic ears; frontal bossing; hypertelorism; Interdigital webbing; Joint hypermobility; long philtrum; metatarsus varus; midface hypoplasia; Shawl scrotum; short stature;Short/Broad Hands; Wide feet; Widow's peak |
| | | NR* | 35(1) | NR* | Anteverted nostrils/Short Nose;Anteverted nostrils/Short Nose; bone age retarded; crease below lower lip; downward slant palpebral fissures; dysplastic ears; Exostosis; hypertelorism; large eyebrows; long philtrum; pectus excavatum; shawl scrotum; short stature; Short/Broad Hands; wide feet; widow's peak |
| | | NR* | 36(4) | NR* | Anteverted nostrils/Short Nose;Brachydactyly/wide fingers; camptodactyly; clinodactyly; clubbed feet; Cryptorchidism; development delayed; Frontal bossing; hypertelorism; interdigital webbing; Shawl Scrotum; Short stature; Short/Broad Hands; Widow's peak |

| | | | | | |
|---|---|---|---|---|---|
| **Pillozzi-Edmonds et al. [28]** | 2011 | CaNA†da | 37(1) | 0.75 | Axial hypotonia; bitemporal NA†rrowing;Clinodactyly; clubbed feet; frontal bossing; hydronephrosis; hypertelorism; inguiNA†l hernia; interphalangeal joint contractures; low set ears; preauricular tag; Shawl scrotum; simian creases; vesicoureteral reflux |
| | | | 38(2) | 0.75 | Anteverted nostrils/Short Nose; Axial Hypotonia; broad NA†sal bridge; clinodactyly; clubbed feet; cryptorchidism; frontal bossing; hypertelorism; interphalangeal joint contracture; long philtrum; low set ears; myopia; preauricular tag; Shawl Scrotum. InguiNA†l Hernia; Simian Creases; thin upper lip vermillion; triangular downturned mouth; upslanted palpebral fissures; vesicoureteral reflux |
| | | | | | |
| **Ronce et al. [29]** | 2012 | France | 39 | NR* | Anteverted nostrils/Short Nose; arthrogryposis;Camptodactyly; cryptorchidism; dysplastic ears; hypertelorism; joint hypermobility; large fontanelle/sutures; long philtrum; low set ears; metatarsus varus; micropenis; midface hypoplasia; RetrogNA†thia; round face; short neck; Short/Broad Hands; simian creases; wide feet; widow's peak |
| | | | | | |
| | | | 40(III-1) | NR* | Anteverted nostrils/Short Nose; cleft lip palate; cow's lick; cryptorchidism; dysplastic ears;Hypertelorism; interphalangeal joint contracture; joint hypermobility; long philtrum; shawl scrotum; short stature; Short/Broad Hands; wide feet |

| | | | | | |
|---|---|---|---|---|---|
| **Aten et al. [30]** | 2012 | Netherlands | 41 (III-2) | NR* | Anteverted nostrils/Short Nose; cow's lick; cryptorchidism; dysplastic ears;Hypertelorism; joint hypermobility; long philtrum; shawl scrotum; short stature; Short/Broad Hands; widow's peak |
| | | | 42 (I-1) | NR* | Cow's lick;Hypertelorism; Interphalangeal joint contractures; long philtrum; Short Stature; Widow's peak |
| | | | | | |
| **Altıncık et al. [31]** | 2013 | Turkey | 43 | 7 | Amblyopia; astigmatism; borderline mental retardation;Brachydactyly/wide fingers; broad NA†sal bridge; cryptorchidism; Frontal bossing; Hypertelorism; interdigital webbing; long philtrum; short stature; Short/Broad Hands |
| | | | | | |
| **Völter et al. [32]** | 2014 | Germany | 44 | 9 | ADHD; Broad NA†sal bridge; cleft lip palate;Clinodactyly; hemianopsia; hyperNA†sal speech; hypertelorism; interdigital webbing; long philtrum; low set ears; metatarsus varus; microcephaly; Midface hypoplasia; NA†sal ectopic teeth; ptosis; round face; short neck; Short stature; Short/Broad Hands; telecanthus; Von Willebrand disease |
| | | | | | |
| | | | 45(Index) | 16 | Anteverted nostrils/Short Nose; Anteverted nostrils/Short Nose; hypertelorism; joint hypermobility; metatarsus varus; MicrogNA†thia; Myopathy; ogival palate; ptosis; shawl scrotum; short stature |

| | | | | | |
|---|---|---|---|---|---|
| **Al-Semari et al. [19]** | 2013 | Saudi Arabia | 46(Bro1‖) | 14 | Anteverted nostrils/Short Nose; Anteverted nostrils/Short Nose; hypertelorism; joint hypermobility; metatarsus varus; microgNA†thia; ogival palate; pectus excavatum; shawl scrotum; short stature |
| | | | 47(Bro2‖) | 21 | Anteverted nostrils/Short Nose; Anteverted nostrils/Short Nose; hypertelorism; joint hypermobility; metatarsus varus; microgNA†thia; ogival palate; shawl scrotum; short stature |
| | | | 48(Nep¶1) | 1 | Anteverted nostrils/Short Nose; Anteverted nostrils/Short Nose; hypertelorism; joint hypermobility; metatarsus varus; MicrogNA†thia; ogival palate; pectus excavatum; ptosis; shawl scrotum; short stature |
| | | | 49(Nep¶2) | 4 | Anteverted nostrils/Short Nose; Anteverted nostrils/Short Nose; coxa magNA†; hypertelorism; joint hypermobility; metatarsus varus; microgNA†thia; myopathy; ogival palate; shawl scrotum; short stature |
| | | | | | |
| **Niida et al.[33]** | 2014 | Japan | 50 | 6 | Anteverted nostrils/Short Nose; brachydactyly/wide fingers; clinodactyly; cryptorchidism; hypertelorism; interdigital webbing; ptosis; round face; shawl scrotum |

| | | | | | |
|---|---|---|---|---|---|
| **Pérez-Coria et al. [4]** | 2015 | Mexico | 51(1) | 6 | Anteverted nostrils/Short Nose; bottom lip fold; brachydactyly/wide fingers; clinodactyly; cryptorchidism; dental malocclusion; dysplastic ears; frontal bossing; hypertelorism; interdigital webbing; joint hypermobility; long philtrum; midface hypoplasia; obesity; shawl scrotum; short stature; simian creases; wide feet; widow's peak |
| | | | 52(4) | 12 | Bottom lip fold; brachydactyly/wide fingers; camptodactyly; Cryptorchidism; Dental malocclusion; downward slant palpebral fissures; dysplastic ears; hypertelorism; interdigital webbing; joint hypermobility; Long Philtrum; Midface Hypoplasia; Obesity; Psychomotor retardation; ptosis; shawl scrotum; Short stature; syndactyly; wide feet; widow's peak |
| | | | 53(5) | 4 | Anteverted nostrils/Short Nose; bottom lip fold; brachydactyly/wide fingers; camptodactyly; clinodactyly; cryptorchidism; downward slant palpebral fissures; dysplastic ears; hypertelorism; inguiNA†l hernia; interdigital webbing; joint hypermobility; long philtrum; midface hypoplasia; ptosis; shawl scrotum; short stature; simian crease; umbilical hernia; wide feet; widow's peak |

| | | | | | |
|---|---|---|---|---|---|
| **GE et al. [20]** | 2015 | ChiNA† | 54 | 2 | Blepharophimosis; Bone age retarded;Brachydactyly/wide fingers; enlarged liver; groove below upper lip; hypertelorism; inguiNA†l hernia; low set ears; short stature; splenic cyst; swollen testicles |
| | | | | | |
| **Parıltay et al. [34]** | 2016 | Turkey | 55 | 14 | Anteverted nostrils/Short Nose;Brachydactyly/wide fingers; broad NA†sal bridge; cryptorchidism; dysplastic ears; hypertelorism; large fontanelle/sutures; ogival palate; shawl scrotum; simian crease; widow's peak |
| | | | | | |
| **Griffin et al. [35]** | 2016 | USA§ | 56(IV-1) | 1 | Ankyloglossia;Anteverted nostrils/Short Nose; astigmatism; brachydactyly/wide fingers; clinodactyly; clubbed feet; cryptorchidism; diastema; downward slant palpebral fissures; dysplastic ears; interphalangeal joint contracture; low set ears; metatarsus varus; ptosis; round face; shawl scrotum; simian creases; still's murmur |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **Hamzeh et al. [36]** | 2017 | UAE‡ | 57(IV-3) | 7 | ADHD;Anteverted nostrils/Short Nose; blepharophimosis; brachydactyly/wide fingers; broad NA†sal bridge; clinodactyly; Development delayed; downward slant palpebral fissures; dysplastic ears; epicanthal fold; frontal bossing; hypertelorism; interdigital webbing; Low set ears; microgNA†thia; ogival palate; pectus cariNA†tum; pectus excavatum; ptosis; short columella; short neck; short philtrum; Short stature; simian creases; sleep disorder; social/language skills delayed; syndactyly; thin upper lip vermillion; widely spaced nipples; widow's peak |
| | | | 58(IV-5) | 3 | ADHD; Anteverted nostrils/Short Nose;Anteverted nostrils/Short Nose; Blepharophimosis; brachydactyly/wide fingers; Broad NA†sal bridge; clinodactyly; Development delayed; downward slant palpebral fissures; dysplastic ears; epicanthal fold; Frontal bossing; hypertelorism; long uvula; Neurocognitive development delayed; overactive frontalis muscle; poor bell's phenomenon; ptosis; short philtrum; short neck; short stature; Short/Broad Hands; telecanthus; widely spaced nipples; widow's peak |

## Legend

*NR: Not reported; †NA: Not applicable; ‡UAE: United Arab Emirates; §USA: Unite

**ɪog-Scott syndrome and assessment of genotype-phenotype correlation: a sys**

ɹsto de Lima Oliveira, Eliene Magda de Assis, Mic hel Campos Ribeiro, Analina Furtado Valadão, A

List of all informations collected from each patient included

| Type of mutation | Splicing Mutation | Amino Acid Change | Protein |
|---|---|---|---|
| Missense/Nonsense (1565G>A) | NA† | Arg522His | p.R522H |
| Missense/Nonsense (1565G>A) | NA† | Arg522His | p.R522H |
| Gross Deletion (Incl.ex.9-12) | NA† | NA† | NA† |

| | | | |
|---|---|---|---|
| Missense/Nonsense (1829G>A) | NA† | Arg610Gln | p.R610Q |
| Missense/Nonsense (1829G>A) | NA† | Arg610Gln | p.R610Q |
| Missense/Nonsense (1829G>A) | NA† | Arg610Gln | p.R610Q |

| | | | |
|---|---|---|---|
| Small Insertion (528insC) | NA† | NA† | NA† |
| Small Insertion (528insC) | NA† | NA† | NA† |
| Missense/Nonsense (614G>T) | NA† | Ser205Ile | p.S205I |

| | | | |
|---|---|---|---|
| Small Deletion (982delC) | NA† | NA† | p.(His328Thrfs*32) |
| Gross Deletions (944-975del32) | NA† | NA† | NA† |
| Missense/Nonsense (1193 A>C) | NA† | Glu380Ala | p.E380A |
| Missense/Nonsense (1328 G>A) | NA† | Arg443His | p.R443H |

| | | | |
|---|---|---|---|
| Small Deletion (1316-1319del AGCT) | NA† | NA† | NA† |
| Small Deletion (2530delG) | NA† | NA† | p.(Val844Trpfs*19) |
| Missense/Nonsense (c.1223G>A) | NA† | Arg408Gln | p.R408Q |

| | | | |
|---|---|---|---|
| Small Deletion (c.2192delA) | NA† | NA† | p.(Lys731Argfs*132) |
| Small Deletion (c.2192delA) | NA† | NA† | p.(Lys731Argfs*132) |
| Small Deletion (c.2192delA) | NA† | NA† | p.(Lys731Argfs*132) |
| Missense/Nonsense (c.1327G>T) | NA† | Arg443Leu | p.R443L |
| Missense/Nonsense (c.2221G>T) | NA† | Glu741Term | p.E741* |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.1396A>G) | NA† | Met466Val | p.M466V |
| | | | |
| Small Insertions (c.944dupC) | NA† | NA† | p.(Ala316Cysfs*4) |
| Small Insertions (c.944dupC) | NA† | NA† | p.(Ala316Cysfs*4) |

| | | | |
|---|---|---|---|
| Gross Deletions (ENTIRE GENE) | NA† | NR* | NR* |
| Small Deletions (806delC) | NA† | NA† | p.(Ala269Valfs*91) |
| Missense/Nonsense (c.1205G>A) | NA† | Arg402Gln | p.R402Q |
| Missense/Nonsense (c.1590T>A) | NA† | Tyr530Term | p.Y530* |

| | | | |
|---|---|---|---|
| Small Deletions (1620delC) | NA† | NA† | Asp540Glufs*11 |
| Missense/Nonsense (c.1673C>G) | NA† | Ser558Trp | p.S558W |
| Splicing (c.1935+3A>C) | IVS11 ds A-C +3 | NA† | NA† |
| Missense/Nonsense (c.1966C>T) | NA† | Arg656Term | p.R656* |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.1966C>T) | NA† | Arg656Term | p.R656* |
| Missense/Nonsense (c.1966C>T) | NA† | Arg656Term | p.R656* |
| Small Deletions (2020_2020 del GAG) | NA† | NA† | Glu676del |
| Missense/Nonsense (c.2242A>G) | NA† | Lys748Glu | p.K748E |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.175C>T) | NA† | Gln59Term | p.Q59* |
| Missense/Nonsense (c.175C>T) | NA† | Gln59Term | p.Q59* |
| Duplication | NA† | NA† | NA† |
| Small deletions (c.2016-35delA) | NA† | NA† | NA† |

| | | | |
|---|---|---|---|
| Small deletions (c.2016-35delA) | NA† | NA† | NA† |
| Small deletions (c.2016-35delA) | NA† | NA† | NA† |
| Splicing (c.482-2A>G) | IVS2 as A-G -2 | NA† | NA† |
| Missense/Nonsense (c.1468C>T) | NA† | Gln490Term | p.Q490* |
| Missense/Nonsense (c.1341G>A) | NA† | Trp447Term | p.W447* |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.1341G>A) | NA† | Trp447Term | p.W447* |
| Missense/Nonsense (c.1341G>A) | NA† | Trp447Term | p.W447* |
| Missense/Nonsense (c.1341G>A) | NA† | Trp447Term | p.W447* |
| Missense/Nonsense (c.1341G>A) | NA† | Trp447Term | p.W447* |
| Missense/Nonsense (c.1340G>A) | NA† | Trp447Term | p.W447* |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.1138G>T) | NA† | Glu380Term | p.E380* |
| Missense/Nonsense (c.1990C>T) | NA† | Gln664Term | p.Gln664* |
| Missense/Nonsense (c.1990C>T) | NA† | Gln664Term | p.Gln664* |

| | | | |
|---|---|---|---|
| Missense/Nonsense (c.1270A>G) | NA† | Asn424Asp | p.N424D |
| Splicing Mutations (c.1340+2T>A) | IVS6 ds T-A +2 | NA† | NA† |
| Missense/Nonsense (c.2761C>T) | NA† | Gly757Arg | p.G757R |

| | | | |
|---|---|---|---|
| Small Deletions (c.53delC) | NA† | NA† | p.(Pro18Argfs*106) |
| Small Deletions (c.53delC) | NA† | NA† | p.(Pro18Argfs*106) |

ed States of America; ||bro: Brother; ¶nep= Nephew

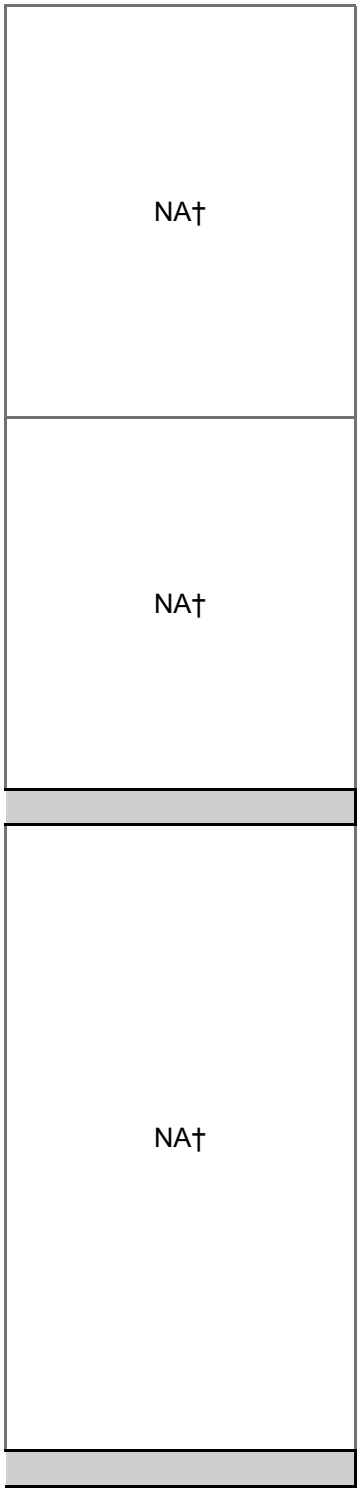
**tematic review**
lfredo Orrico

| Insertions/Duplications |
| --- |
| NA† |
| NA† |
| NA† |

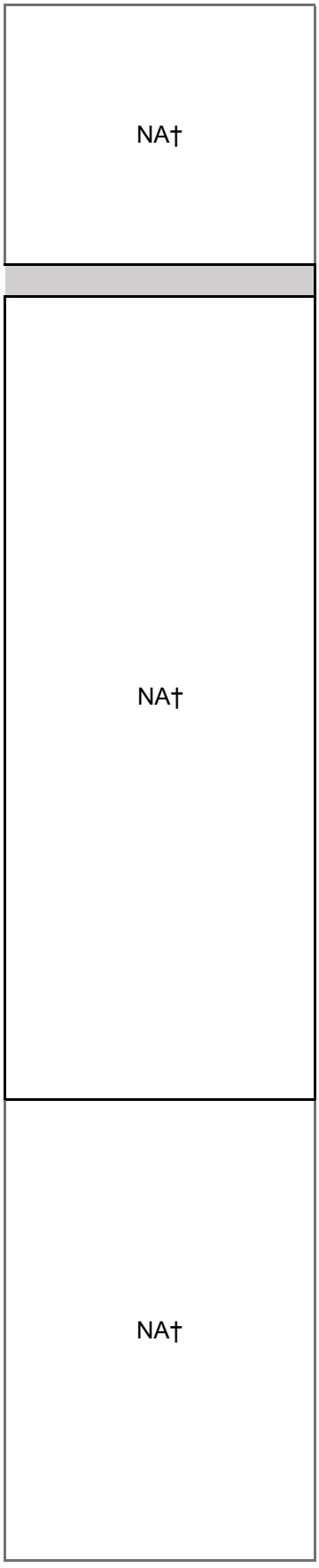
| |
|---|
| NA† |
| NA† |
| NA† |

| |
|---|
| NA† |
| NA† |
| NA† |

| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |


NA†
NA†
NA†

| |
|---|
| NA† |
| NA† |
| NA† |
| |
| NA† |
| NA† |
| |


NA†
NA†
NA†

| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |

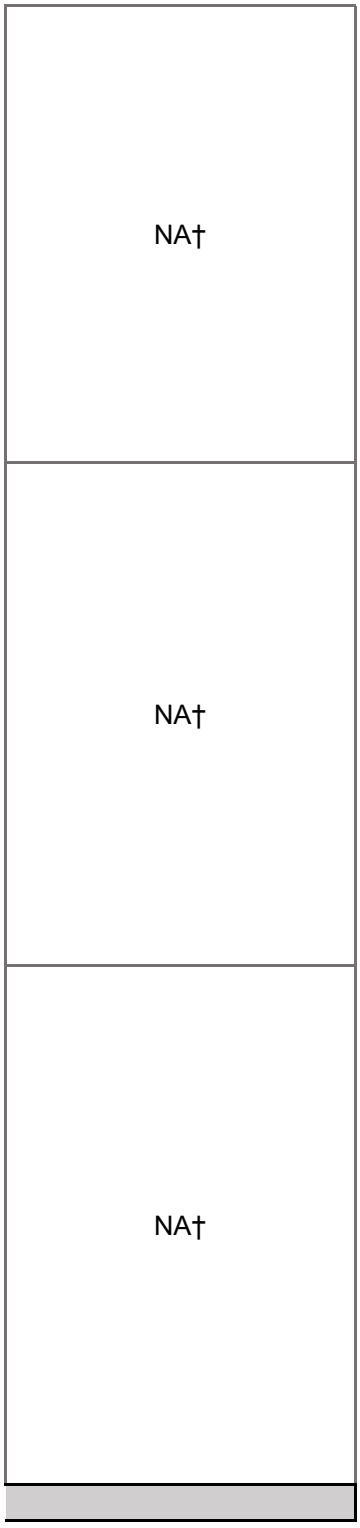
| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |

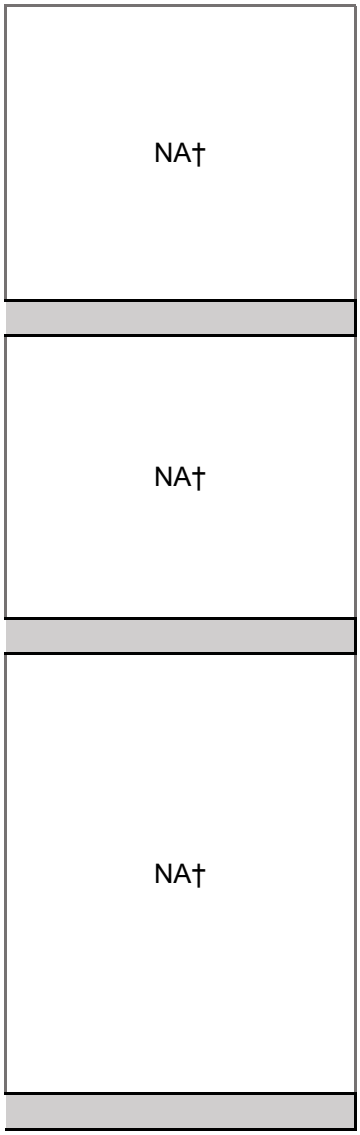
| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |
| |

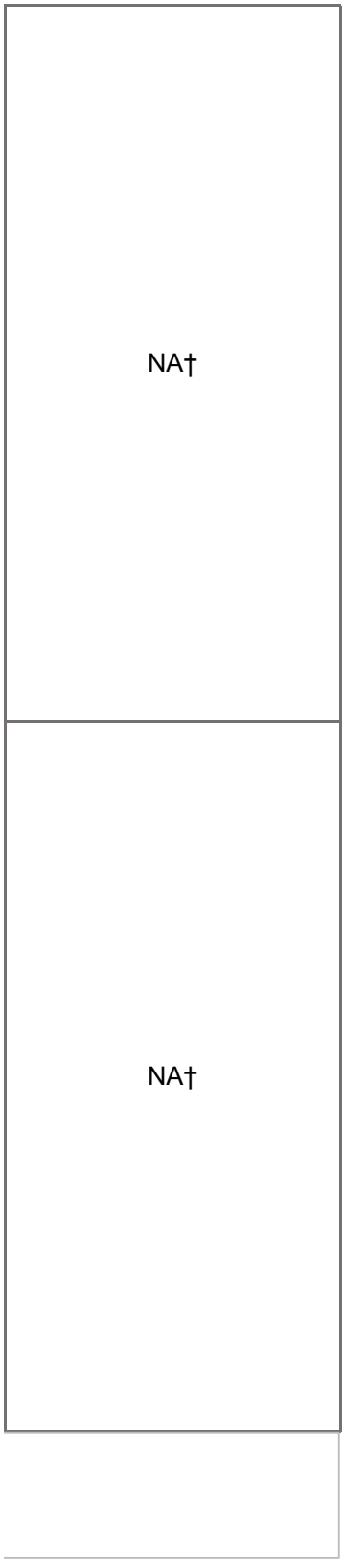
| |
|---|
| NA† |
| NA† |
| c.307+5711_2016-227 |
| NA† |

| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |
| NA† |

| |
|---|
| NA† |
| NA† |
| NA† |
| NA† |
| |
| NA† |
| |


NA†
NA†
NA†


NA†
NA†
NA†


NA†
NA†